B3.2 Transport (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

8 MARKS

1 **DEFINE** tissue fluid. [B3.2.11] [1]

2 **STATE** one component of plasma that is almost absent from tissue fluid. [B3.2.12] [1]

3 **NAME** the valve between the left atrium and the left ventricle. [B3.2.15] [1]

4 **STATE** where the sinoatrial node is found. [B3.2.16] [1]

5 **OUTLINE** two features of lymph vessels. [B3.2.13] [2]

6 **STATE** two differences between the heart of a bony fish and the heart of a mammal. [B3.2.14] [2]

Section B: Understanding

16 MARKS

7 **EXPLAIN** how tissue fluid is formed and reabsorbed along a capillary. [B3.2.11] [3]

8 **EXPLAIN** the advantage of a double circulation over a single circulation. [B3.2.14] [3]

9 **EXPLAIN** why the wall of the left ventricle is thicker than the wall of the right ventricle. [B3.2.15] [2]

10 **EXPLAIN** how substances are exchanged between tissue fluid and the cells of a tissue. [B3.2.12] [3]

11 **EXPLAIN** how cardiac muscle is adapted to the work of the heart. [B3.2.15] [3]

12 **EXPLAIN** the importance of the delay at the atrioventricular node. [B3.2.16] [2]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

13 [B3.2.16]

Pressures were recorded in the left side of the heart of a person at rest during one cardiac cycle of 0.8 s.

TIME / S	0	0.1	0.2	0.3	0.4	0.5	0.6
LEFT ATRIUM / MMHG	6	9	5	7	10	5	6
LEFT VENTRICLE / MMHG	5	8	70	118	90	4	5
AORTA / MMHG	80	77	75	116	100	90	85

a **STATE** the systolic and diastolic pressures in the aorta. [2]

b **CALCULATE** the pulse pressure. [1]

c **IDENTIFY** one time when the atrioventricular valve is open. [1]

d **IDENTIFY** the time at which both valves are shut. [1]

e **DEDUCE**, with a reason, whether the semilunar valve is open at 0.3 s. [2]

f **CALCULATE** the heart rate. [1]

g **EXPLAIN** why the pressure in the aorta stays high while the ventricle is relaxed. [2]

Section D: Longer answer

6 MARKS

14 **DESCRIBE** the events in the left side of the heart during one cardiac cycle, starting with the firing of the sinoatrial node. [B3.2.16] [6]